

Yi (Reid) Chen

Ph.D. Candidate in Electrical and Computer Engineering, University of Wisconsin–Madison

Expected graduation: 2028 | Advisor: Ramya Korlakai Vinayak

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Research Overview

Reliable, compute-efficient factuality verification for LLMs: Conformal factuality and risk control for RAG under distribution shift; masked-diffusion and lightweight entailment verifiers; agent evaluation with LLM and VLM judges. Related threads: confidence scores for auto-labeling, human feedback, and pluralistic alignment.

Publication record: Seven conference papers at NeurIPS, ICML, ICLR, WWW (oral), AISTATS, UAI, and HCOMP; first or co-first author on three (ICML, WWW, HCOMP); plus a first-author 2026 preprint.

Research Experience

Conformal Factuality for RAG-based LLMs: Robustness and Efficiency Analysis 2025–2026

- *arXiv 2026 (first author); ICML Workshop 2025:* Led a study across three benchmarks and multiple model families; found that lightweight entailment verifiers match or outperform LLM confidence scorers while requiring over $100\times$ fewer FLOPs.
- Introduced informativeness-aware metrics; diagnosed vacuous outputs at high factuality levels and the fragility of conformal filtering under distribution shifts and distractors.

Palimpsest: Bidirectional Scoring for Span-Level Hallucination Detection 2026–present

- Proposed a masked-diffusion verifier that masks entity spans and scores length-normalized token recovery; designed reference-grounded and reference-free evaluations against trained checkers and token-probability baselines.
- Designed a matched cloze comparison of a masked-diffusion language model (Dream-7B) and an autoregressive LLM (Qwen2.5-7B) to isolate reconstruction-based verification behavior; ongoing work extends the approach to conformal factuality under generator shift.

Confidence Scores for Reliable Data Annotation and Pseudolabeling 2024–2025

- *NeurIPS 2024:* Co-developed confidence functions and thresholding methods for automated labeling under explicit error constraints, improving coverage by up to 60% over calibration baselines while holding error below 5%.
- *ICML 2025 (equal contribution):* Connected automated-label quality and quantity through a coverage–error objective in pseudolabeling-based semi-supervised learning.

Human Feedback, Preference Learning, and Pluralistic Alignment 2023–2025

- *ICLR 2025:* Co-developed PAL, a pluralistic alignment framework that learns shared structure and individual differences from heterogeneous preferences.
- *WWW 2025 (oral):* Designed crowdsourced clustering studies of cognitive overload and contextual bias to understand how question format changes annotation quality.
- *AISTATS 2024; UAI 2025:* Studied identifiability and recovery of preference distributions from limited pairwise comparisons, and nonlinear metric learning in an RKHS, respectively.
- *HCOMP 2023:* Developed active querying for crowdsourced clustering with theoretical recovery guarantees and evaluation on human-generated data.

Industry Experience

Meta — Core Ads Growth Engineering

2026

Software Engineer Intern, Machine Learning (Summer 2026)

Part-time Student Researcher (Fall 2026)

- **Ad Creative Agent (Jun–Jul 2026):** Designed and shipped a production multi-turn, multi-model agent for generating, editing, and iterating on ad creatives from advertiser briefs.
- Architected a ReAct orchestrator with seven tools and four specialized advisors, integrating diffusion-based generation and editing, a vision-language critic, and a learned performance scorer.
- Developed supervised fine-tuning and model-serving infrastructure; built a VLM-as-judge evaluation suite with metrics, dashboards, and mock-based tests.

Education

University of Wisconsin–Madison

Ph.D., Electrical and Computer Engineering

2023–present (expected 2028)

M.S., Electrical and Computer Engineering

2022–2025

B.A., Computer Sciences, *Honors in the Major*

2019–2022

B.A., Mathematics

2019–2022

Publications

* denotes equal contribution.

Preprint

- [1] **Chen, Y.**, Chen, D., Chikodikar, S. M., Yin, C. H., and Vinayak, R. K. “Is Conformal Factuality for RAG-based LLMs Robust? Novel Metrics and Systematic Insights.” *arXiv:2603.16817*, 2026.

Conference Publications

- [2] Tatli, G., **Chen, Y.**, Mason, B., Nowak, R. D., and Vinayak, R. K. “Metric Learning in an RKHS.” *Uncertainty in Artificial Intelligence (UAI)*, 2025.
- [3] Vishwakarma, H.*, **Chen, Y.***, Namburi, S. S. S. GNVV, Tay, S. J., Vinayak, R. K., and Sala, F. “Rethinking Confidence Scores and Thresholds in Pseudolabeling-based SSL.” *International Conference on Machine Learning (ICML)*, 2025.
- [4] **Chen, Y.** and Vinayak, R. K. “Query Design for Crowdsourced Clustering: Effect of Cognitive Overload and Contextual Bias.” *The ACM Web Conference (WWW)*, 2025. **Oral presentation.**
- [5] Chen, D., **Chen, Y.**, Rege, A., Wang, Z., and Vinayak, R. K. “PAL: Pluralistic Alignment Framework for Learning from Heterogeneous Preferences.” *International Conference on Learning Representations (ICLR)*, 2025.
- [6] Vishwakarma, H., **Chen, Y.**, Tay, S. J., Namburi, S. S. S., Sala, F., and Vinayak, R. K. “Pearls from Pebbles: Improved Confidence Functions for Auto-labeling.” *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- [7] Tatli, G., **Chen, Y.**, and Vinayak, R. K. “Learning Populations of Preferences via Pairwise Comparison Queries.” *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024.
- [8] **Chen, Y.**, Vinayak, R. K., and Hassibi, B. “Crowdsourced Clustering via Active Querying: Practical Algorithm with Theoretical Guarantees.” *AAAI Conference on Human Computation and Crowdsourcing (HCOMP)*, 2023.

Workshop Publications

- [9] **Chen, Y.**, Yin, C. H., Chikodikar, S. M., and Vinayak, R. K. "On the Scoring Functions for RAG-based Conformal Factuality." *ICML Workshop on Reliable and Responsible Foundation Models*, 2025.
- [10] Vishwakarma, H., **Chen, Y.**, Namburi, S. S. S. GNVV, Tay, S. J., Vinayak, R. K., and Sala, F. "PabLO: Improving Semi-Supervised Learning with Pseudolabeling Optimization." *ICML Workshop on Data-Centric Machine Learning Research*, 2024.
- [11] **Chen, Y.** and Vinayak, R. K. "Query Design for Crowdsourced Clustering: Effect of Cognitive Overload and Contextual Bias." *ICML Workshop on Models of Human Feedback for AI Alignment*, 2024.
- [12] Tatli, G., **Chen, Y.**, and Vinayak, R. K. "Learning Populations of Preferences via Pairwise Comparison Queries." *ICML MFPL Workshop*, 2023.
- [13] **Chen, Y.**, Vinayak, R. K., and Hassibi, B. "Crowdsourced Clustering via Active Querying: Practical Algorithm with Theoretical Guarantees." *ICML AI & HCI Workshop*, 2023.
- [14] Tatli, G., **Chen, Y.**, and Vinayak, R. K. "Learning Preference Distributions From Pairwise Comparisons." *International Workshop on Computational Social Choice (COMSOC)*, 2023.

Public Research Artifacts

Open-source software: PAL implementation and datasets (Apache-2.0; ICLR 2025).

Public code and data: WWW 2025 (Zenodo), AISTATS 2024, and HCOMP 2023.

Teaching and Mentoring

University of Wisconsin–Madison

Summer 2025, Fall 2025, Fall 2026

Teaching Assistant, ECE 539: Introduction to Artificial Neural Networks

- Mentored undergraduate researchers on research projects and scientific writing.

Professional Service

Conference reviewer: NeurIPS 2026; COLM 2026; UAI 2026; ICLR 2025, 2026; WWW 2025.

Workshop reviewer: NeurIPS 2024; ICML 2024.

Awards and Scholarships

HCOMP/CI Travel Award, 2023

UW-Madison Conference Presentation Award

UW-Madison Undergraduate Scholarship for Summer Study

James Graaskamp Scholarship

Technical Skills

Machine learning: PyTorch, JAX, Hugging Face Transformers, TensorFlow, scikit-learn; supervised fine-tuning and evaluation.

Programming: Python (primary), C/C++, CUDA, Java, JavaScript, Rust, SQL.

Research and evaluation: LLM factuality verification and conformal prediction; agent evaluation with LLM and VLM judges; human feedback and preference learning.

Systems: Model serving, Docker, Git, AWS (EC2, S3, Lambda).